

Course 4061

4 Credits: 4

Program core

Source-grounded

Yarn Manufacture II

□ To impart the knowledge of constructional features and working of Speed frame,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4061 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4061.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	4061
Course title	Yarn Manufacture II
Semester	4 Credits: 4
Course category	Program core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the knowledge of constructional features and working of Speed frame, Ring frame, doubling machines & advanced spinning systems
- To understand the process involved in the conversion of sliver to yarn
- To estimate the draft, production and efficiency of the machines.

Course outcomes

CO	Official outcome	Study level
CO1	18 Applying frame. Describe the process of yarn formation through	See official syllabus
CO2	condensed yarn spinning and other advanced 15 Understanding spinning techniques. Discuss the different methods of yarn doubling	See official syllabus
CO3	10 Understanding techniques and reeling of yarn. Illustrate and solve the problems related to draft,	See official syllabus
CO4	twist, angle of yarn pull, traveler lag and 15 Applying determine production of various machines.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 1
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Illustrate the working of Speed frame and Ring frame	4	As specified
Module 2	and advanced spinning techniques.	4	As specified
Module 3	of yarn.	4	As specified
Module 4	determine production of speed frame, ring frame, rotor spinning, Ring	4	As specified

MODULE 1

Illustrate the working of Speed frame and Ring frame

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the working of Speed frame and Ring frame

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

5 Understanding the functions of the parts involved. Discuss the modern developments in speed

5 Understanding the functions of the parts involved. Discuss the modern developments in speed is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding the functions of the parts involved. Discuss the modern developments in speed using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding the functions of the parts involved. discuss the modern developments in speed should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding the functions of the parts involved. Discuss the modern developments in speed means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-5 Understanding the functions of the parts involved. Discuss the modern developments in speed $\frac{\text{distance}}{\text{time}}$ definition/meaning $\frac{\text{ദൂരം}}{\text{സമയം}}$. $\frac{\text{distance}}{\text{time}}$ $\frac{\text{ദൂരം}}{\text{സമയം}}$ $\frac{\text{distance}}{\text{time}}$, steps, application $\frac{\text{distance}}{\text{time}}$ $\frac{\text{ദൂരം}}{\text{സമയം}}$ $\frac{\text{distance}}{\text{time}}$. Technical terms English- $\frac{\text{distance}}{\text{time}}$ $\frac{\text{ദൂരം}}{\text{സമയം}}$.

4 Understanding frame. Describe the working of Ring frame and the

4 Understanding frame. Describe the working of Ring frame and the is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding frame. Describe the working of Ring frame and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding frame. describe the working of ring frame and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding frame. Describe the working of Ring frame and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Understanding frame. Describe the working of Ring frame and the definition/meaning.
 steps, application. Technical terms English-

functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame

functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions of the parts involved and building 5 understanding mechanism of ring frame. identify and analyse the possible ring frame should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഈ functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഏകദേശം ഏകദേശം.

processing defects and suggest remedies to 4 Applying solve them.
Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system.
Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning

processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning is an

official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, processing defects and suggest remedies to 4 applying solve them. objectives of speed frame - working of speed frame machine- skf pk 1500 drafting system - constructional details of flyer assembly - function of cots, condenser, false twister, spindle, principles of winding - flyer lead and bobbin lead, objectives of builder motion, objectives of differential motion, developments in speed frame -

features of multi motor drive system, roving stop motion using light barriers, auto doffing and bobbin transport system. objectives of ring frame - working of ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, traversing thread guide, abc ring, separator, top roller cots and shore hardness, aprons-rings types - low crown ring & anti-wedge ring - comparison of c-traveller and elliptical traveler - concept of traveller number and ring flange number ,running in period of ring, construction details of modern spindle, twisting principle, objectives of building mechanism, working of building mechanism of ring frame for weft wind, methods for achieving variable spindle speed in spinning - vfd & vps. identify and analyze the possible ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. describe the process of yarn formation through condensed yarn spinning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons- Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning definition/meaning , steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Illustrate the working of Speed frame and Ring frame		
Explain the working of simplex machine and		
M1.01		5
Understanding	the functions of the parts involved.	
	Discuss the modern developments in speed	
M1.02		4
Understanding	frame.	
	Describe the working of Ring frame and the	
M1.03	functions of the parts involved and building	5
Understanding	mechanism of Ring frame.	
	Identify and analyse the possible Ring frame	
M1.04	processing defects and suggest remedies to	4
Applying	solve them.	
Contents:		
Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false		
twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder		
motion, Objectives of Differential motion, Developments in speed frame - Features of multi		
motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

and advanced spinning techniques.

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and advanced spinning techniques.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and

compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compact spinning and benefits of compact 3 understanding yarn discuss the principle of open end spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and compact spinning definition/meaning. Compact spinning steps, application of compact spinning. Technical terms English-compact spinning.

4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and

4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of rotor spinning. illustrate the principle of friction spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഘട്ടം 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and ഏകദേശം നിർവ്വചനം definition/meaning നിർവ്വചനം. ഏകദേശം നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം. Technical terms English-ഘട്ടം നിർവ്വചനം നിർവ്വചനം.

4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and

4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of dref ii spinning machine. explain the principle of air jet spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and its definition/meaning. Also, explain its steps, application and technical terms in English.

4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling

4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor

diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of murata air jet spinner. need for compacting element in condensed yarn spinning - concept of spinning geometry - working of elite compact spinning system - advantages of compact yarn - working of siro spinning-different advanced spinning systems - principle of open end spinning system- working of rotor spinning system - functions of opening roller, transport tube, rotor, navel, take up tube - relation between fibre length and rotor diameter, rotor speed - types of rotor groove - advantages and limitations of rotor spinning over ring spinning - principle of friction spinning - working of dref ii friction spinning - principle of air jet spinning - fasciated yarn - working of murata air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. describe the different mechanism of yarn doubling techniques and reeling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

and advanced spinning techniques.

M2.01	Explain the spinning geometry, mechanism of compact spinning and benefits of compact yarn	3
Understanding		

M2.02	Discuss the principle of open end spinning and working of Rotor spinning.	4
Understanding		

M2.03	Illustrate the principle of Friction spinning and working of DREF II spinning machine.	4
Understanding		

M2.04	Explain the principle of Air jet spinning and working of Murata Air jet spinner.	4
Understanding		

	Series Test - I	1
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Contents:

Need for Compacting element in condensed yarn spinning - concept of spinning Geometry -

Working of Elite compact spinning system - advantages of compact yarn - Working of

SIRO spinning-Different advanced spinning systems - principle of open end spinning

system- working of Rotor spinning system - Functions of opening roller, transport

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

of yarn.

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of yarn.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding doubling Illustrate the working of Ring doubler and

2 Understanding doubling Illustrate the working of Ring doubler and is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding doubling Illustrate the working of Ring doubler and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding doubling illustrate the working of ring doubler and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding doubling Illustrate the working of Ring doubler and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding doubling Illustrate the working of Ring doubler and റിംഗ് ഡബ്ലർ definition/meaning റിംഗ് ഡബ്ലർ. റിംഗ് ഡബ്ലർ റിംഗ് ഡബ്ലർ, steps, application റിംഗ് ഡബ്ലർ റിംഗ് ഡബ്ലർ. Technical terms English- റിംഗ് ഡബ്ലർ റിംഗ് ഡബ്ലർ.

3 Understanding TFO machine Discuss the types of fancy yarn and working of

3 Understanding TFO machine Discuss the types of fancy yarn and working of is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding TFO machine Discuss the types of fancy yarn and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding tfo machine discuss the types of fancy yarn and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding TFO machine Discuss the types of fancy yarn and working of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-3 Understanding TFO machine Discuss the types of fancy yarn and working of ഫാൻസി യാർൻ definition/meaning ഫാൻസി യാർൻ. ഫാൻസി യാർൻ ഫാൻസി, steps, application ഫാൻസി ഫാൻസി. Technical terms English- ഫാൻസി ഫാൻസി.

3 Understanding fancy doubler. Describe the objectives of reeling and the

3 Understanding fancy doubler. Describe the objectives of reeling and the is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding fancy doubler. Describe the objectives of reeling and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding fancy doubler. describe the objectives of reeling and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding fancy doubler. Describe the objectives of reeling and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 3 Understanding fancy doubler. Describe the objectives of reeling and the റീലിംഗ് ഡബ്ബിംഗ് definition/meaning റീലിംഗ് ഡബ്ബിംഗ്. റീലിംഗ് ഡബ്ബിംഗ് റീലിംഗ്, steps, application റീലിംഗ് ഡബ്ബിംഗ് റീലിംഗ്. Technical terms English- റീലിംഗ് ഡബ്ബിംഗ്.

2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and

2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the

concept of draft, twist, angle of yarn pull, traveler lag and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding concept of bundling and baling. contents objects of doubling - dry and wet doubling - english and scotch system of wet doubling - effect of twist direction - d/s ratio of doubled yarn - types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - types of doubling machine - working of ring doubler - objectives of assembly winder - working of tfo - merits of tfo over ring doubler - define fancy yarn - types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - working of fancy doubler - objectives of reeling - working of plain reeling machine - concept of bundling and baling - relationship between number of knots in a bundle and count. illustrate the concept of draft, twist, angle of yarn pull, traveler lag and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch

system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and
 definition/meaning . steps, application . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

of yarn.

M3.01	Discuss the objectives and the methods of doubling	2
Understanding		
M3.02	Illustrate the working of Ring doubler and TFO machine	3
Understanding		
M3.03	Discuss the types of fancy yarn and working of fancy doubler.	3
Understanding		
M3.04	Describe the objectives of reeling and the concept of bundling and baling.	2
Understanding		

Contents

Objects of doubling - dry and wet doubling - English and scotch system of wet doubling -
 Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler -
 Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

determine production of speed frame, ring frame, rotor spinning, Ring

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: determine production of speed frame, ring frame, rotor spinning, Ring
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn

4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying and machine efficiency of speed frame determine the tpi, actual draft, angle of yarn should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn ഏകദേശം നിർണ്ണയിക്കുക. definition/meaning നിർണ്ണയിക്കുക. ഏകദേശം നിർണ്ണയിക്കുക, steps, application നിർണ്ണയിക്കുക നിർണ്ണയിക്കുക. Technical terms English- 4 നിർണ്ണയിക്കുക നിർണ്ണയിക്കുക.

pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine

pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pull, traveller speed, traveller lag, and twist 4 applying contraction in ring frame. determine the production and machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പുല, ട്രാവലർ സ്പീഡ്, ട്രാവലർ ലാഗ്, and twist 4 Applying contraction in Ring frame. Determine the production and machine definition/meaning. പൂർണ്ണമായും നിർവ്വചിക്കുക. നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and

efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, efficiency of ring frame and rotor spinning 4 applying machine. determine the resultant count , production and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഈ efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഈ ഏകദേശം ഏകദേശം.

efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com

efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production

& efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, efficiency of ring doubler and tfo 3 applying machine roving hank calculation - tpi and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of rotor spinning machine - resultant count - production and efficiency of ring doubler & tfo machine. text / reference: t/r book title/author t1 w. klein, rieter manual of spinning volume 4-6 , rieter, 2010 r. v. m. gowda, new spinning systems, ncute publication, new delhi, r2 2005 r3 oxtoby e., "spun yarn technology", butterworths, london, 2002. r4 dr.a.r.khare- elements of ring frame and doubling chattopadhyay r, "advances in technology of yarn production", ncute, iit r5 delhi, 2002. online resources: sl.no website link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഏകദേശം efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com ഏകദേശം efficiency definition/meaning ഏകദേശം efficiency definition/meaning, steps, application ഏകദേശം efficiency definition/meaning. Technical terms English-ഏകദേശം efficiency definition/meaning.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

determine production of speed frame, ring frame, rotor spinning, Ring doubler and TF0 machine.		
Determine the TPI, actual draft, production		
M4.01		4
Applying		
and machine efficiency of speed frame		
Determine the TPI, actual draft, angle of yarn		
M4.02		4
Applying		
pull, traveller speed, traveller lag, and twist		
contraction in Ring frame.		
Determine the production and machine		
M4.03		4
Applying		
efficiency of Ring frame and Rotor spinning		
machine.		
Determine the resultant count , production and		
M4.04		3
Applying		
efficiency of Ring doubler and TF0		
machine		
Series Test - II		1
Contents:		
Roving hank calculation - TPI and actual draft calculation of speed frame & ring		
frame -		
concept of angle of yarn pull - traveler speed - traveller lag - twist		
contraction and its		
calculation - production and efficiency of speed frame & ring frame - production		
&		
efficiency of Rotor spinning machine - Resultant count - production and		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 5 Understanding the functions of the parts involved. Discuss the modern developments in speed. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding the functions of the parts involved*. Discuss the modern developments in speed. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding frame. Describe the working of Ring frame and the. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding frame*. Describe the working of Ring frame and the. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame. (3 marks)

Answer: Begin with the standard definition or identity of *functions of the parts involved and building 5 Understanding mechanism of Ring frame*. Identify and analyse the possible Ring frame. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction

system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning. (3 marks)

Answer: Begin with the standard definition or identity of *processing defects* and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: തുടങ്ങി സ്റ്റാൻഡർ ഡിഫിനിഷൻ, പ്രശ്നങ്ങൾക്ക് പരിഹാരം, 4 പ്രയോഗിക്കുക പരിഹരിക്കുക. സ്പീഡ് ഫ്രെയിം - സ്പീഡ് ഫ്രെയിം മെഷീൻ- SKF PK 1500 ഡ്രാഫ്റ്റിംഗ് സിസ്റ്റം - ഫ്ലൈർ അസംബ്ലി - കോട്ട്, കണ്ടൻസർ, ഫോൾസ് ട്വിസ്റ്റർ, സ്പിൻഡിൾ, വൈൻഡിംഗ് - ഫ്ലൈർ ലീഡ്, ബോബിൻ ലീഡ്, ബിൽഡർ മൂവ്മെന്റിന്റെ ലക്ഷ്യം, ഡിഫറൻഷ്യൽ മൂവ്മെന്റിന്റെ ലക്ഷ്യം, സ്പീഡ് ഫ്രെയിമിലെ വികാസങ്ങൾ - മൾട്ടി മോട്ടർ ഡ്രൈവ് സിസ്റ്റം, റോവിംഗ് സ്റ്റോപ്പ് മൂവ്മെന്റ് ഉപയോഗിച്ച് ലൈറ്റ് ബാരിയറുകൾ, ഓട്ടോ ഡോഫ്റ്റിംഗ്, ബോബിൻ ട്രാൻസ്പോർട്ട് സിസ്റ്റം. റിംഗ് ഫ്രെയിം - റിംഗ് ഫ്രെയിം മെഷീൻ - LR പ്നിയുമറ്റിക് ടോപ്പ് അം ഡ്രാഫ്റ്റിംഗ് സിസ്റ്റം- ബോബിൻ ഹോൾഡർ, റോവിംഗ് ട്രാവേഴ്സ് ഗൈഡ്, ഫ്ലൂട്ടഡ് ആൻഡ് ക്നൂർലഡ് റോളർ, സ്പേസർ, ക്രാഡിൾ, നോസ് ബാർ, റോളർ ക്ലിയറർ, പ്നിയുഫൈൽ സൂക്ഷ്യാവകാശ്യം, ട്രാവേഴ്സിംഗ് ത്രേഡ് ഗൈഡ്, ABC റിംഗ്, സെപറേറ്റർ, ടോപ്പ് റോളർ കോട്ട്, ഷോർ ഹാർഡ്നസ്സ്, ഏപ്രോൺ-റിംഗ്സ് ടൈപ്പ് - ലോ ക്രൗൺ റിംഗ് & ആന്റി-വെഡ്ജ് റിംഗ് - C-ട്രാവെല്ലർ, എലിപ്റ്റിക്കൽ ട്രാവെല്ലർ - ട്രാവെല്ലർ നമ്പർ, റിംഗ് ഫ്ലാഞ്ച് നമ്പർ, റിംഗ് ഓൺ ടൈം, റിംഗ് ഫ്രെയിമിന്റെ നിർമ്മാണ വിവരങ്ങൾ, മോഡേൺ സ്പിൻഡിൾ, ട്വിസ്റ്റിംഗ് പ്രിൻസിപ്പിൾ, ബിൽഡിംഗ് മെക്കാനിസത്തിന്റെ ലക്ഷ്യം, ബിൽഡിംഗ് മെക്കാനിസത്തിന്റെ റിംഗ് ഫ്രെയിം ഫോർ വേഫ്റ്റ് വിൻഡ്, വേരിയബിൾ സ്പിൻഡിൾ സ്പീഡ് ഇൻ സ്പിന്നിംഗ് - VFD & VPS. ഐഡൻറിഫൈ ആൻഡ് അനലൈസ് ചെയ്യുക റിംഗ് ഫ്രെയിം പ്രോസസ്സിംഗ് ഡിഫക്റ്റുകൾ, ഉദാഹരണം: എക്സസ്സീവ് എൻഡ് ബ്രേക്കുകൾ, സ്ലബ്ബുകൾ, തുടർച്ചയായ ഓർഡിനറി, സോഫ്റ്റ് യാർൺ, നെപ്സ്, സ്മാർൽസ്, പരിഹാരം നിർദ്ദേശിക്കുക. ഡിസ്കൈസ് ചെയ്യുക യാർൺ ഫോർമേഷൻ ടൂർണിംഗ് കണ്ടൻസ്ഡ് യാർൺ സ്പിന്നിംഗ്. സ്റ്റേറ്റ് ഗവേർണിംഗ് പ്രിൻസിപ്പിൾ, ലിസ്റ്റ് ചെയ്യുക കീ കമ്പോൺറന്റ്സ്, ഓൺ ഓൺ റിലവന്റ അപ്ലിക്കേഷൻ, ഓൺ ഓൺ ഓർഗനൈസേഷൻ. ഉപയോഗിക്കുക ഔദ്യോഗിക ടെർമിനോളജി, ഓൺ ഓൺ ഓർഗനൈസേഷൻ.

Module 2

Define or explain compact spinning and benefits of compact 3 Understanding yarn Discuss the principle of open end spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *compact spinning and benefits of compact 3 Understanding yarn* Discuss the principle of open end spinning and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രവർത്തനം/ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവ ഉൾപ്പെടെയുള്ള വിവരങ്ങൾ.

Define or explain 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രവർത്തനം/ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവ ഉൾപ്പെടെയുള്ള വിവരങ്ങൾ.

Define or explain 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രവർത്തനം/ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവ ഉൾപ്പെടെയുള്ള വിവരങ്ങൾ.

Define or explain 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system -

Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding doubling Illustrate the working of Ring doubler and. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding doubling Illustrate the working of Ring doubler and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding TFO machine Discuss the types of fancy yarn and working of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding TFO machine* Discuss the types of fancy yarn and working of. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും definition, principle/steps, application എന്നിവ ഉൾപ്പെടെ ഉപയോഗിക്കുക.

Define or explain 3 Understanding fancy doubler. Describe the objectives of reeling and the. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding fancy doubler. Describe the objectives of reeling and the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും definition, principle/steps, application എന്നിവ ഉൾപ്പെടെ ഉപയോഗിക്കുക.

Define or explain 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling*

machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine. (3 marks)

Answer: Begin with the standard definition or identity of *pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and. (3 marks)

Answer: Begin with the standard definition or identity of *efficiency of Ring frame and Rotor spinning 4 Applying machine*. Determine the resultant count , production and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com. (3 marks)

Answer: Begin with the standard definition or identity of *efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine*. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 <https://nptel.ac.in> 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com. State the governing principle, list the key components or stages, and close with one relevant

application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **5 Understanding the functions of the parts involved. Discuss the modern developments in speed.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding doubling Illustrate the working of Ring doubler and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://nptel.ac.in> www.reiter.com www.laksmimach.com
www.suessen.com www.fibre2fashion.com

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 4061. Verify institutional updates against the current official curriculum.